

Principles of Economics

Class 9: Banks, Money and the Credit Market

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- ▶ So far: **firms and consumers interact in markets** to their mutual benefit;
- ▶ That interaction needs money **as a medium of exchange**.
- ▶ Today, the **role of money and banks**:
 - ▶ Why do individuals **lend, save and invest**?
 - ▶ How does the **banking system shape consumption** and the economy?
 - ▶ Are there **risks** associated with banking activity?
 - ▶ What do **commercial banks and the central bank** do?
 - ▶ How do banks **create money**?
 - ▶ What are the **limitations** of the banking system?

Plan for today's class

Introduction

Saving and borrowing model

Balance sheet

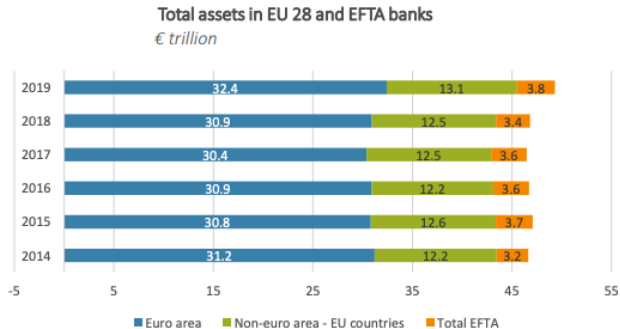
Banking as a business

Banks and money

The money and credit market

Conclusion

1. Introduction



In 2020, EU

GDP was €13.3 trillion (English)

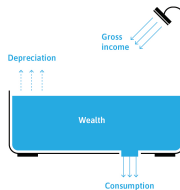
The financial system and economic crises

País / Región	Periodo	Desencadenante principal	Qué falló en el sistema bancario	Impacto económico principal
EE. UU. (Gran Depresión)	1930–1933	Pánicos bancarios tras el crack del 29	Bancos poco capitalizados, sin seguro de depósitos, colapso de colaterales	PIB se hunde, desempleo = 25%, fuerte deflación
Japón	1990s–2000s	Estallido de burbuja inmobiliaria y bursátil	Grandes préstamos incobrables, ocultamiento de pérdidas, bancos 'zombi'	Estancamiento del PIB y repetidas recesiones
Suecia y Finlandia (crisis nórdica)	1991–1993	Burbuja inmobiliaria y liberalización financiera	Exceso de crédito y riesgo inmobiliario	Fuerte recesión, posterior recuperación tras saneamiento
Indonesia (Crisis Asiática)	1997–1998	Devaluación y salida de capitales	Endeudamiento externo en divisas, falta de supervisión	PIB cae > 10%, pobreza y desempleo se disparan
Tailandia / Corea del Sur	1997–1998	Crisis del baht y contagio regional	Préstamos en dólares, burbuja inmobiliaria	Contracción severa del PIB y desempleo elevado
Rusia	1998	Default soberano y devaluación	Exposición bancaria a deuda pública y divisas	Recesión abrupta y pérdida de confianza
Argentina	2001–2002	Crisis del régimen de convertibilidad	Mismatches de moneda y plazo, corrida bancaria	Profunda recesión, desempleo y pobreza muy altos
Islandia	2008–2010	Colapso de grandes bancos globalizados	Bancos sobredimensionados y apalancados en divisas	PIB cae con fuerza, desempleo temporal elevado
España	2008–2013	Estallido de burbuja inmobiliaria	Exposición de cajas a créditos promotores, mala gestión de riesgos	Larga recesión y desempleo masivo
Chipre	2012–2013	Pérdidas en deuda griega y sobreexposición bancaria	Altas posiciones en bonos soberanos y crédito inmobiliario	Contracción del PIB y control de capitales
Grecia	2015	Crisis soberana y control de capitales	Interdependencia bancos-Estado, depósitos fugados	Agravamiento de la recesión y del desempleo

Definitions

- ▶ **Wealth** is the stock of goods owned or the value of that stock:
$$[\text{Wealth}] = [\text{Money, land, buildings, machinery, capital goods, etc.}]$$
 - [Debts you owe]
 - + [Debts owed to you]
- ▶ **Gross income** (a flow): money received after taxes from labour, the market, investments or subsidies.
- ▶ **Net income**: the maximum amount one can consume without reducing one's wealth.
$$[\text{Net income}] = [\text{gross income after taxes}] - [\text{depreciation}]$$

(**Depreciation** is the reduction in the value of wealth over time.)
- ▶ **Savings**: net income not consumed;
- ▶ **Investment**: spending on capital goods (gross/net depending on whether depreciation is included).



2. Saving and borrowing model

- ▶ A **trade-off between consuming now or later**.
- ▶ Finance and lending move purchasing power across time:
 - ▶ **Borrow, today**: more income to consume now.
 - ▶ **Borrow, future**: less income (repayment + interest) to consume later.
 - ▶ **Lend or save, today**: less income to consume now.
 - ▶ **Lend or save, future**: more income (funds + interest) to consume later.
- ▶ Whether and how much to borrow or lend **depends on**:
 - ▶ Income today, and expected income (and inflation) tomorrow;
 - ▶ The interest rate offered;
 - ▶ Consumers' preferences.
- ▶ The **interest rate** is the **price** of moving money across time.

Modelling borrowing

The feasible set

- ▶ A two-period borrowing model.
- ▶ Julia has no income today (t) but $E_{t+1} = 100\$$ tomorrow (t+1)

- ▶ **Without borrowing:** zero today, \$100 tomorrow.

- ▶ **Borrowing:** consume today at the expense of tomorrow:

$$C_t = B_t \text{ and}$$

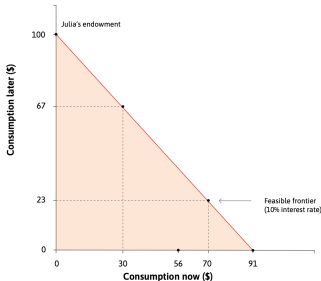
$$C_{t+1} = B_t(1 + r) = E_{t+1}$$

Therefore,

$$C_{t+1} = E_{t+1} - (1 + r)C_t$$

Where C: Consumption;

E: Endowment; r: Interest rate



Possible combinations of current and future consumption, given the interest rates.

- ▶ **Opportunity cost** of consuming tomorrow: consuming today.
- ▶ $(1 + r)$ is the cost of transforming consumption today into future consumption (MRT)

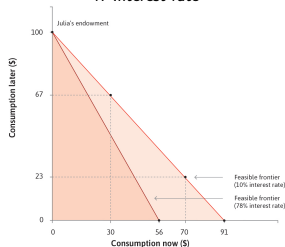
Modelling borrowing

The feasible set

- ▶ With an **interest rate of $r = 10\%$** :
 - ▶ At most \$91 today.
 - ▶ Tomorrow she repays \$91 + interest = \$100.
- ▶ But what if the **interest rate were $r = 78\%$** ?
 - ▶ At most \$56 today.
 - ▶ And repays \$56 + interest = \$100.

$$C_{t+1} = E_{t+1} - (1+r)C_t$$

Where C: Consumption; E: Endowment;
r: Interest rate

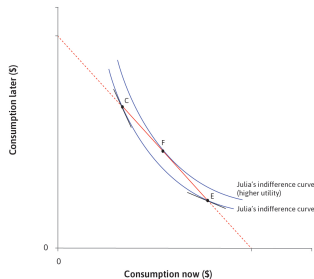


Possible combinations of current and future consumption, given the interest rates.

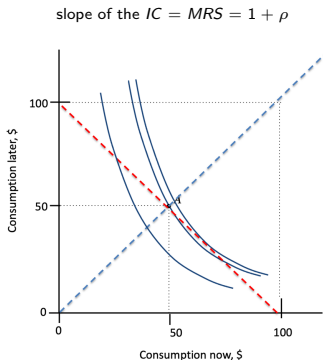
- ▶ A **higher interest rate shrinks the feasible set**: consuming today costs more.

- ▶ Preferences between now and later rest on two things:
 - ▶ **Consumption smoothing** (avoiding large changes in consumption over time)
 - ▶ **Impatience** (preferring to consume now)

- ▶ **Consumption smoothing** is diminishing marginal returns to consumption:
- ▶ Better two similar periods than one rich and one poor.
- ▶ C means consuming little today and a lot tomorrow.
- ▶ E means consuming a lot today and little tomorrow.
- ▶ **Point F provides the highest utility.**

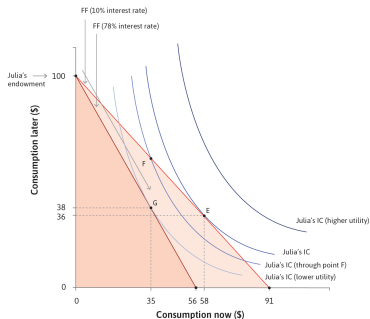


- ▶ But consuming **now** beats consuming later.
- ▶ The slope is **pure impatience**:
 - ▶ **Myopia**: present satisfaction is worth more than future satisfaction;
 - ▶ **Prudence**: future consumption is uncertain.

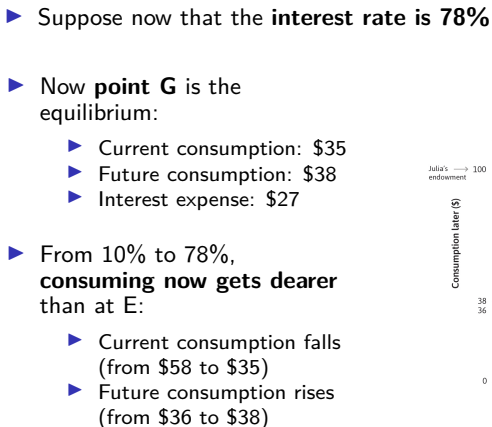


- ▶ Impatience is measured by the **discount rate** (ρ):
 - ▶ If ρ is high, current consumption is valued more (consumer impatience, preference for present consumption)
 - ▶ If ρ is low, future consumption is valued more (consumption smoothing, balance between now and the future)

- ▶ **Equilibrium:** the feasible combination of consumption now and later with the highest utility.
- ▶ Geometrically it is the **tangency point** where:
 $MRS = MRT \Leftrightarrow 1 + \rho = 1 + r \Leftrightarrow \rho = r$
- ▶ If the interest rate is **10%**, **point E** is the equilibrium:
 - ▶ Current consumption \$58
 - ▶ Future consumption \$36
 - ▶ Interest $\$58 \times 0.10 \approx \6
- ▶ **Point F is not an equilibrium:**
 $1 + \rho > 1 + r \Leftrightarrow \rho > r$
Therefore, it is better to consume more now.



Equilibrium – changes in interest rates



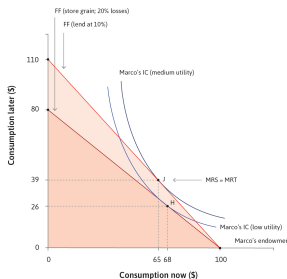
- ▶ The **study of saving is similar**.

$$C_t + A_t = E_t \text{ and } C_t = (1 + r)A_t$$

$$\text{Therefore, } C_{t+1} = (1 + r)(E_t - C_t)$$

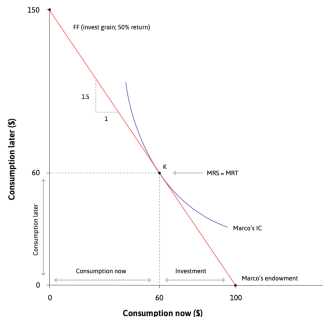
- ▶ Marco has **\$100 now and expects nothing** in $t + 1$.
- ▶ Unlike Julia, Marco can:
 - ▶ **Consume \$100 now and nothing later,**
 - ▶ **Or save and lend part of it,** and consume the repayment plus interest later.
 - ▶ Interest rate: 10%;
 - ▶ The loan is risk-free.
- ▶ **Saving smooths consumption by postponing it.**

- ▶ Marco **sells his grain today, consumes part and lends the rest**:
- ▶ He would earn **10% interest**:
 - ▶ If he lends \$100 today, he can consume \$110 tomorrow.
 - ▶ Lending takes him **above \$100**.
- ▶ With the larger feasible set and his preferences:
 - ▶ **J would be the equilibrium point**, where:
he consumes \$65 now and \$39 later;
- ▶ Note that if Marco **consumes \$65 now**:
he lends $\$100 - \$65 = \$35$;
in the future he gets $\$35 + \text{interest}$:
 $\$35 \times (1 + 0.1) \approx \39
- ▶ Lending beats storing the grain.



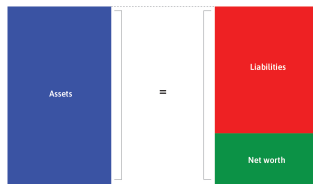
- ▶ The same model serves for **investment decisions**.
- ▶ Suppose Marco could **invest \$100** in:
 - ▶ **Better technology**: \$150 of grain in the future.
 - ▶ This investment has a **50% return**;

- ▶ The frontier **shifts out** along the vertical axis.
- ▶ The **equilibrium** would now occur at **point K** where:
 - ▶ He consumes \$60 today and invests \$40;
 - ▶ The \$40 turn into \$60 next period.



3. Balance sheet

- ▶ A **balance sheet**: what you own and what you owe.
 - ▶ **Asset**: anything of value that is owned
 - ▶ **Liability**: anything of value that is owed
 - ▶ **Net worth** = **Assets** - **Liabilities** = **Equity**



- ▶ **Assets = Liabilities + Net worth:**
 - ▶ If $\text{Assets} > \text{Liabilities}$, $\text{Net worth} > 0 \Rightarrow$ you own more than you owe;
 - ▶ If $\text{Assets} < \text{Liabilities}$, $\text{Net worth} < 0 \Rightarrow$ you owe more than you own (**Insolvency**).

- ▶ Net worth **does not change when you lend or borrow:**

- ▶ A loan adds an asset and a liability:

- ▶ The **loan money** (liquidity) is an **asset**;
- ▶ The **debt** is the corresponding **liability**:
that money has to be repaid.

- ▶ Julia's case, from earlier:

- ▶ Julia faces a 10% interest rate:
- ▶ She **borrow**s \$58 to consume now;
- ▶ In the future, she **will have to repay** \$58 + \$6 (interest) = \$64
- ▶ She consumes \$36 at the end: net worth 0.

	JULIA'S ASSETS		JULIA'S LIABILITIES	
Now (before consuming)	Cash	\$58	Loan	\$58
			Net worth = \$58 - \$58	0
	JULIA'S ASSETS		JULIA'S LIABILITIES	
Now (after consuming)	Cash	0	Loan	\$58
			Net worth	-\$58
	JULIA'S ASSETS		JULIA'S LIABILITIES	
Later (before consuming)	Cash	\$100	Loan	\$64
			Net worth = \$100 - \$36	\$36
	JULIA'S ASSETS		JULIA'S LIABILITIES	
Later (after consuming)	Cash	\$64	Loan	\$64
			Net worth	0

4. Banking as a business

- ▶ A bank is a firm that profits from borrowing and lending:
 - ▶ **Banks' costs:**
 - ▶ Interest on their liabilities: deposits and other loans
 - ▶ Operating costs: salaries, branches
 - ▶ **Banks' revenues:**
 - ▶ Interest and repayment of loans.
 - ▶ Fees, and other products such as insurance
- ▶ The profit is the spread: they charge more than they pay.
- ▶ And they carry the risk of default.

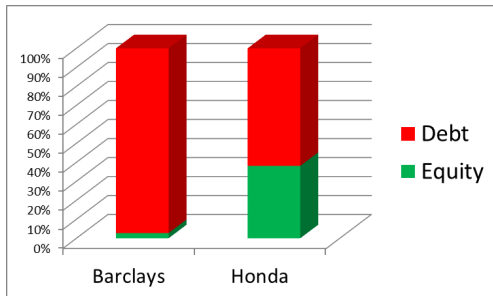
Banks' balance sheet

► **A bank's balance sheet** is not like a household's:

- **Assets:** bank loans (money out)
- **Liabilities:** bank borrowing (money in)

Activos (propiedad del banco o adeudado a éste)		% del balance	Pasivos (lo que el banco debe a los hogares, empresas y otros bancos)		% del balance
1. Saldos de efectivo y reservas en el banco central	Propiedad del banco: fondos de acceso inmediato	2	1. Depósitos	Propiedad de hogares y empresas	50
2. Activos financieros, algunos de los cuales (bonos del Estado) pueden utilizarse como garantía para obtener préstamos	Propiedad del banco	30	2. Préstamo garantizado (se proporciona garantía)	Incluye préstamos de otros bancos a través del mercado monetario	30
3. Préstamos a otros bancos	A través del mercado monetario	11	3. Préstamo no garantizado (sin garantía)		16
4. Préstamos a hogares y empresas (por ejemplo, hipotecas)		55			
5. Activos fijos, como edificios y equipos	Propiedad del banco	2			
Total activos		100	Total pasivos		96
			4. Patrimonio neto = Total de activos – total de pasivos = capital propio		4

- ▶ A bank's **balance sheet** is different from a normal firm's:



- ▶ **Leverage** = Debt/Equity
- ▶ Banks usually have very high leverage.

- ▶ **Banks' net worth is determined by:**

Net worth = Total assets - total liabilities = Own capital

- ▶ Net worth is what is owed to the owners, also called **capital**.
- ▶ **Negative net worth:** more liabilities than assets, the bank is **insolvent**.
- ▶ One **financial ratio** above all: **leverage**.

$$\text{Leverage} = \frac{\text{Total Assets}}{\text{Net Worth}}$$

- ▶ A **high ratio** means heavy reliance on debt, and a riskier bank.

Defaults, liquidity risk and banking crises

- ▶ Taking deposits and making loans forces banks into:
 - ▶ **Maturity transformation:**
 - ▶ Deposits: short term;
 - ▶ Loans: long term.
 - ▶ **Liquidity transformation:**
 - ▶ Deposits are liquid;
 - ▶ Loans are not liquid.
- ▶ This exposes banks to **two types of risk**:
 - ▶ **Default risk:** customers do not repay.
 - ▶ **Liquidity risk:** not enough cash to return deposits.

Bank run in Germany, 1931



- ▶ In a **bank run** every depositor asks at once, and the bank can fail.
- ▶ Banks can also fail due to **bad investments** (default risk);
- ▶ Governments are likely to intervene because of **systemic risk**.

5. Banks and money

- ▶ A **bank** profits from borrowing and lending.

- ▶ **Two types of bank:**

1. **Central bank: a sovereign entity** that issues currency, regulates the banking system and sets monetary policy.

- ▶ In the EU this is done mainly by the **European Central Bank**;
- ▶ National central banks (the Bank of Spain) keep regulatory duties and follow the ECB's guidelines.

2. **Commercial banks:** banks that interact with other economic agents (consumers and firms). **Video**.

- ▶ Like any firm, they are **for-profit**;
- ▶ They **take deposits**, borrow from the central bank or other banks, and **lend**.
- ▶ Their revenue is the **gap between the deposit rate and the lending rate**. The interest they pay on deposits is generally lower than on loans → profit.

- ▶ Central banks are responsible for the **monetary base / high-powered money**:
 - ▶ **Cash in circulation**: coins and notes;
 - ▶ **Legal tender**: money that must be accepted as payment by law.
- ▶ The central bank is the **only bank that can create legal tender**:
 - ▶ A **government body**, independent of the government or not.
- ▶ The central bank acts as a **banker for commercial banks**:
 - ▶ **Commercial banks hold accounts at the central bank** with legal tender;
 - ▶ With these balances, **the central bank can create money**:
 - ▶ It prints money and injects it as **liquidity to commercial banks**.

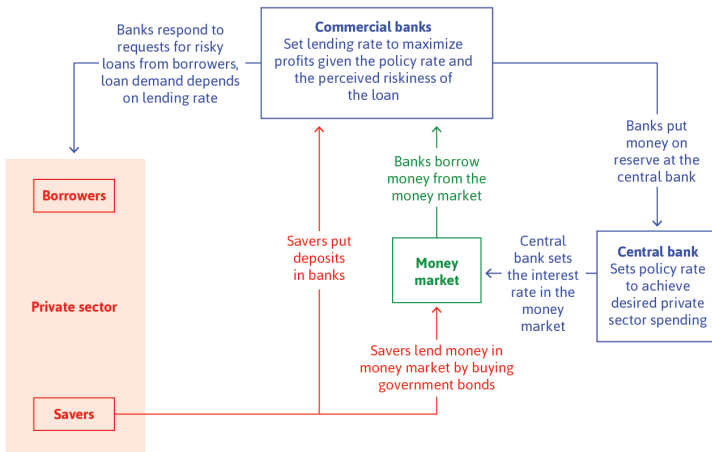
- ▶ **Commercial banks create money** by making loans: [Video](#)
 - ▶ This is bank money, NOT legal tender
 - ▶ **The loans provided** by commercial banks **generate a liability** for the bank!
 - ▶ They **profit** from the interest on that bank money.
- ▶ They create money by **lending far more than they hold in deposits**.
- ▶ Gino has \$20 at Bonus Bank and **borrow \$100 from it**.
- ▶ The **loan** is recorded as an asset, but also as a **liability**:
Now Gino can consume \$120!
- ▶ This **increases the bank's balance sheet**:
Now, total assets are \$120, the bank "creates" \$100!
- ▶ The bank's balance sheet would be as follows:
 - ▶ Bonus Bank's assets: \$20 base money + \$100 the bank lends to Gino = Total \$120
 - ▶ Bonus Bank's liabilities: \$120 payable on demand to Gino
- ▶ In general, **money in the economy** includes base money and bank money:
Money = Base money + Bank money

6. The money and credit market

- ▶ Banks **need base money** to settle their net transactions, and they get it in the money market:
 - ▶ Commercial banks borrow base money at the **policy interest rate**.
- ▶ **Demand for base money** follows the volume of transactions:
 - ▶ *Positive relationship* with economic growth and the price level;
 - ▶ *Negative* with interest rates: saving means not using money.
- ▶ **The supply of base money** is decided by the central bank:
 - ▶ It is part of monetary policy decisions.
- ▶ **Two different types of interest rates**:
 - ▶ **Policy rate**: set by the central bank, applied to base money.
 - ▶ **Bank lending rate**: what commercial banks charge firms and households.
- ▶ **The policy rate influences the lending rate**, but it is not the only factor.

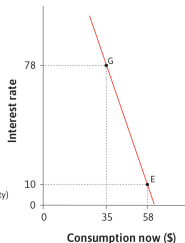
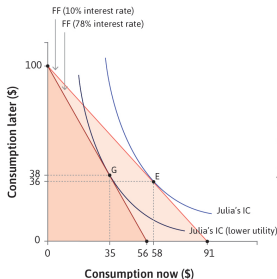
The financial system

- The **financial system**: commercial banks between households and firms on one side, and the central bank on the other.



The policy interest rate and the central bank

- ▶ Through the interest rate, **the central bank** moves the whole banking system and, with it, **the economy**:



- ▶ Careful: **the central bank does not control what commercial banks charge**;
- ▶ In general, they are free to set their own rates;
- ▶ But the **policy rate** makes the money market dearer and **raises banks' costs**.
- ▶ And Julia, we saw, consumes less today when the rate is higher.

Central banks and the economy

The policy interest rate and the central bank

European Central Bank announces new super hike of interest rates to tame inflation COMMENTS

By [Jorge Liboreiro](#) • Updated: 28/10/2022



European Central Bank (ECB) President Christine Lagarde speaks during a news conference following the ECB's monetary policy meeting in Frankfurt, Germany October 27, 2022. - Copyright: REUTERS-Wolfgang Rattay

- ▶ The ECB has recently raised its policy interest rate to fight inflation;
- ▶ The aim: cool activity to slow price rises.
- ▶ Read the [article](#).
- ▶ And this [NYTimes article](#) on whether this is a good idea or not.

- ▶ **The banking system is full of asymmetric information.**
 - ▶ **Principal-agent problem:** a conflict over hidden actions or attributes that no contract can pin down.
- ▶ Think about **financing a project**:
 - ▶ The lender risks not being repaid;
 - ▶ And cannot see **the project's prospects or the borrower's effort**;
 - ▶ No contract can guarantee that the project succeeds.
- ▶ This problem can be **mitigated in three ways**:
 - ▶ **Equity:** the borrower puts in part of their own wealth (the deposit on a house);
 - ▶ **Collateral:** property that passes to the lender if the loan is not repaid;
 - ▶ **Personal guarantee:** a third party answers for the debt.

- ▶ And the three of them are **harder to provide with less wealth**.
 - ▶ **Credit rationing**: those with less wealth,
 - ▶ *borrow with more difficulty* than the wealthy (credit-constrained)
 - ▶ are *denied loans* entirely (excluded from credit)
- ▶ Inequality grows when some can profit by lending to others.
- ▶ **Credit rationing increases inequality**:
 - ▶ Those with little wealth cannot take the investment opportunities open to the rest.

7. Conclusion

Section by section (Micro, Unit 9 and Macro, Unit 6)

1. **Introduction** → wealth is a stock and income a flow; saving is net income not consumed.
2. **Saving and borrowing model** → the interest rate is the price of moving money across time; equilibrium where $\rho = r$.
3. **Balance sheet** → assets = liabilities + net worth; borrowing adds both sides and leaves net worth unchanged.
4. **Banking as a business** → profit is the spread; maturity and liquidity transformation bring default and liquidity risk.
5. **Banks and money** → the central bank creates base money, commercial banks create bank money by lending.
6. **The money and credit market** → the policy rate influences, but does not fix, the bank lending rate.
7. **Credit rationing** → equity, collateral and guarantees are harder with less wealth: credit rationing feeds inequality.

Key concepts: wealth, gross and net income, depreciation, saving, investment, discount rate, consumption smoothing, impatience, feasible frontier, balance sheet, net worth, leverage, maturity and liquidity transformation, bank run, base money, bank money, policy interest rate, credit rationing.